

CSCE 671 HUMAN-COMPUTER INTERACTION · MODULE 12 · LECTURE NOTES

Manipulative Design

The same techniques, aimed at the user.

MODULE OUTCOMES

- Recognise the standard deceptive patterns.
- Explain the mechanism each one exploits.
- Distinguish persuasion from manipulation.
- Explain the professional position and its limits.
- Review a design for manipulative patterns.

1 The patterns

Pattern	What it does	The mechanism it exploits
Confirmshaming	Phrases the decline option so that choosing it feels shameful.	Social discomfort — 'No thanks, I don't want to save money'.
Roach motel	Easy to enter and disproportionately hard to leave.	Asymmetric tolerance for effort (\$2's last point).
Preselected opt-in	Obtains consent by default rather than by choice.	Default dominance (Module 05 §3) — the most effective of all of these.
Visual misdirection	Emphasises the option the designer wants chosen.	Preattentive capture (Module 02 §3).
Forced continuity	Begins charging silently when a free trial ends.	Forgetting and inattention (Module 03).
Manufactured urgency	False scarcity, countdown timers, 'three others are viewing this'.	Loss aversion, which is a strong and well-measured effect.
Obstruction	Many steps required in one direction and one step in the other.	Effort asymmetry — see the note.

THESE ARE THE COURSE'S OWN FINDINGS, APPLIED AGAINST THE USER

Default dominance (Module 05 §3), **preattentive capture** (Module 02 §3), **limited attention** (Module 03 §1), and **the non-reading finding** (Module 01 §2) — **all of them are exploitable, and all of them are being exploited commercially at scale.**

Which means the knowledge in this course is dual-use — there is no version of it that helps you design well and does not also help you manipulate — and pretending otherwise would be dishonest, which is why this module is in the course rather than omitted from it.

So the distinction has to lie in the aim rather than in the technique — which is §3's subject and is where the only usable line can be drawn, since the techniques are identical.

And recognising a pattern is most of resisting it, which is **precisely why they are catalogued and named: a named pattern is arguable in a design review,** where an unnamed discomfort is not and loses to a metric.

2 Why they work

- **Defaults are not changed** (Module 05 §3) — so **a preselected checkbox is consent from most people,** and **it is the most effective pattern in the table** precisely because it exploits the strongest finding in the course.
- **Emphasis captures attention preattentively** (Module 02 §3) — so **making one button prominent and the other grey decides the outcome without any false statement being made,** which is what makes it hard to object to on factual grounds.
- **Attention is finite and interruptions are costly** (Module 03 §1) — so **a request made at a moment of focus is accepted simply to be rid of it,** which is why the timing of a prompt is part of the pattern.
- **Nobody reads** (Module 01 §2) — so **disclosure in prose is disclosure that did not happen,** and a design can be simultaneously fully disclosed and fully deceptive.
- **And effort asymmetry works: people abandon a ten-step cancellation and complete a one-step signup. The effort asymmetry is the one that needs no deception at all — every individual step is honest, and the aggregate is a design that prevents people from leaving,** which is why it survives legal review and is the pattern to watch for most closely.

3 Persuasion and manipulation

PERSUASION ADDRESSES THE PERSON'S JUDGEMENT; MANIPULATION BYPASSES IT

Telling somebody a true and relevant fact so that they can decide better is **persuasion** — and a good default chosen for their benefit and visibly changeable is **persuasion too**, since it informs rather than circumvents.

Exploiting a known cognitive limitation in order to produce a choice they would not make on reflection is **manipulation** — and the test is whether the technique survives being explained to them.

Which is the operative test: would you be comfortable telling the user exactly how this works? 'We made the decline button grey because people click the prominent one' fails it immediately and obviously, whereas 'we preselected the safe option' does not.

And it is a usable test precisely because it does not require agreeing on a theory of autonomy — it only requires imagining the explanation, which anybody can do in a design review and which does not depend on anybody's philosophical commitments. That makes it the rare ethical criterion that is actually operable in a meeting.

- A safe default the user would not have chosen — which is **paternalistic and is frequently right** (CSCE 701 Module 11 §3's security defaults), and which passes the explainability test easily.
- Friction deliberately added to a destructive action, which is **obstruction aimed at the user's own benefit** — and is exactly the same technique as the roach motel, differing only in whose interest it serves.
- Genuine scarcity, honestly displayed, which is **indistinguishable at a glance from the manufactured kind** — and which the manufactured kind has made less credible for everybody, a collective cost.
- And a design that is simply *better* at what it does, which **increases engagement without any manipulative intent at all** — so engagement is not itself evidence of anything.
- So the test is the aim and the explainability, not the technique — which is why a list of forbidden techniques is not the answer: every one of them has a legitimate use, and a rule banning friction would forbid confirming a deletion.

4 The professional position

YOU WILL BE ASKED TO BUILD THESE, AND THE USEFUL RESPONSE IS A MEASUREMENT

The request usually arrives as a metric rather than as a design: 'increase signups', 'reduce churn', 'improve consent rates' — and the manipulative implementation is frequently the cheapest available way to move it, which is why it gets built.

So the effective counter is measuring the cost rather than objecting on principle: the support load, the refund and chargeback rate, the complaint rate, the app-store reviews, and the retention of users who felt tricked.

Because manipulated conversions are worse conversions — a subscription somebody did not mean to start is cancelled, charged back, or complained about — and those costs land elsewhere in the organisation, which is precisely why they are invisible to the person asking for the metric. Making them visible is the intervention.

And where the measurement does not win, say so plainly and let the decision be made explicitly — which is a much smaller thing than refusing and is usually available: a recorded objection and a named decision-maker is a real outcome, and it is the one CSCE 701 Module 01 §2's accepted-risk practice would recommend.

- **Compare the effort required in the two directions** — signing up against cancelling, opting in against opting out, adding against removing — and count the steps for each.
- **Check the visual weight of the options**, and ask **whether the recommended one is recommended for the user's benefit** or for somebody else's (§3's test).
- **Check what is preselected**, and **whether anybody chose it deliberately** or it is simply whatever the framework defaults to (Module 05 §3) — both happen, and the second is fixable without an argument.
- **Apply the explainability test to each persuasive element** (§3), one at a time rather than to the design as a whole.
- **And read the whole flow as somebody who wants to leave, which is the perspective nobody designs for** and which no other review covers. **Reading the flow as somebody who wants to leave is the review nobody runs** — and it finds the obstruction patterns immediately, because they were built on the assumption that nobody would look.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Brignull — Deceptive Patterns (free)	§1's catalogue — the patterns named, with examples and the enforcement history.
Mathur et al. — Dark Patterns at Scale (free)	§1 measured — an automated survey of eleven thousand shopping sites, with prevalence figures.

Resource	What it covers for this module
Thaler & Sunstein — Nudge, and its critics	§3's difficult cases — the case for benign paternalism, and the objections to it. Library copy.
Susser, Roessler & Nissenbaum — Online Manipulation (free)	§3's distinction, argued properly — manipulation as bypassing rather than engaging deliberation.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Find each of the seven patterns** in software you use.
- 2 **For each, name the mechanism** and the earlier module it came from.
- 3 **Count the steps to cancel** three subscriptions you hold.
- 4 **Compare against the steps to start them.**
- 5 **Find a preselected checkbox** and determine whether anybody chose it.
- 6 **Apply the explainability test** to five persuasive elements in your own work.
- 7 **Find a legitimate use of obstruction** and justify it.
- 8 **Estimate the support cost** of one manipulative pattern you have seen.
- 9 **Read one of your own flows as somebody trying to leave**, and record what you find.
- 10 **Write the measurement** you would bring to a request for a manipulative pattern.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name seven patterns and the mechanism each exploits.
- 2 Where did each mechanism appear earlier in the course?
- 3 Why is this knowledge dual-use, and what follows?
- 4 Why is effort asymmetry the one needing no deception?
- 5 State the persuasion/manipulation distinction and the operative test.
- 6 Why is the test usable in a design review?
- 7 Give four genuinely difficult cases.
- 8 Why is a list of forbidden techniques not the answer?
- 9 What is the effective professional response, and why does it work?

10 Give five review checks, and the one nobody runs.